

Homework Set 1

Translation-invariance on a lattice

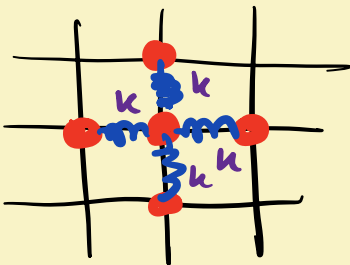
1. a) Consider the following generalization of the model we discussed in class:

$$H = \sum_i \frac{p_i^2}{2m} + \frac{1}{2} \sum_{ij} \lambda_{ij} q_i q_j$$

where λ_{ij} depends only on " $i-j$ ".

- Show that the model is translationally invariant.
- Find the normal modes.

- b) Consider the case the index " i " runs over a two-dimensional lattice and K_{ij} is non-zero only for nearest neighbors & is independent of location:
- $$\lambda_{ij} = \delta_{j(i\pm 1)} \lambda$$



- Find the normal modes
- Find the dispersion relation.

One-dimensional Heisenberg model:

- 2) Consider a one-dimensional array of ferromagnets with $\vec{S}_i$ a vector degree of freedom on each site. Take the Hamiltonian

$$H = -J \sum_i \vec{S}_i \cdot \vec{S}_{i+1}$$

with $J > 0$. The operators $\vec{S}_i$ satisfy the $SU(2)$ algebra

$$[(\vec{S}_i)_\alpha, (\vec{S}_j)_\beta] = i \delta_{ij} \epsilon^{\alpha\beta\gamma} (\vec{S}_j)_\gamma$$

- a) Write down the classical e.o.m for $\vec{S}_i$ variables as

$$\frac{d}{dt} \vec{S}_i = \dots$$

- b) Treat spins as classical variables and take the continuum limit to find the e.o.m of the low energy field.